



UTS

TEXTILE-BASED MOTION SENSING FOR THE HAND (STROKE REHAB)

Handover Plan

TextileTech

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1. Project Overview

Project Title

Textile-Based Motion Sensing Glove for Stroke Rehabilitation

Project Purpose

The purpose of this project is to develop a wearable sensor-integrated rehabilitation glove capable of monitoring finger movement and fingertip force during hand rehabilitation exercises. The system aims to support stroke patients by providing real-time rehabilitation data while improving comfort, portability, and usability through textile-integrated sensing technology.

Problem Being Solved

Many stroke patients experience reduced rehabilitation progress after returning home from supervised clinical rehabilitation programs. Although rehabilitation is most effective during the first three to six months after a stroke, patients often struggle with maintaining exercise consistency due to limited supervision, low motivation, incorrect exercise execution, and difficulties accessing physiotherapy services. Existing rehabilitation technologies may also be expensive, uncomfortable, difficult to use, or unable to accurately capture hand movement.

This project addresses these challenges by developing a lightweight wearable rehabilitation glove capable of tracking finger motion and fingertip force in a home rehabilitation environment.

Target Users

The primary target users are stroke patients in the subacute and chronic stages of recovery who experience hand motor impairments such as reduced grip strength, limited finger movement, and loss of fine motor control. The system is intended to support users performing rehabilitation exercises independently in home-based environments.

High-Level Solution

The proposed solution is a sensor-integrated rehabilitation glove that combines flex sensors, force sensing resistors (FSRs), conductive textile yarn, and an ESP32 microcontroller to monitor hand movement and force application in real time. The glove is designed to measure finger joint angles and fingertip pressure while transmitting sensor data to external software for visualization and rehabilitation monitoring. The project progressively transitions from a conventional wired sports glove prototype toward a textile-integrated wearable design using conductive yarn traces to improve flexibility, comfort, and wearability.

2. Project Progress

During the first six weeks of the project, preliminary research was conducted to support the ideation and design phase of the rehabilitation glove system. The research focused on stroke rehabilitation biomechanics, hand anatomy, rehabilitation stages, sensor technologies, and wearable rehabilitation systems. These findings helped the team understand stakeholder needs, rehabilitation challenges, and the technical feasibility of different sensing approaches. The following sections summarise the key findings from this research phase.

Completed

- Research into stroke rehabilitation, biomechanics, hand anatomy, and rehabilitation stages
- Investigation and evaluation of sensor technologies including flex sensors, FSR sensors, and conductive yarn
- Initial concept design and system architecture development
- Independent testing and calibration of flex sensors and FSR sensors
- Development of the alpha prototype using a sports glove with conventional wiring
- Testing of sensor functionality, accuracy, signal acquisition, and glove integration on the sports glove prototype
- Integration of conductive yarn traces into the glove structure
- Development and refinement of the beta textile-integrated glove prototype
- Testing conductive yarn performance on single finger and multiple finger gloves, durability, and electrical connectivity
- Improving sensor pocket design, glove fitting, and overall wearability

Future Iterations

- Full system testing using the conductive yarn integrated glove prototype
- Long-term user testing with stroke patients
- Gamification and rehabilitation feedback features
- Mobile or advanced software application development
- Clinical validation and large-scale rehabilitation testing

2.1 Research in Stroke rehabilitation

A stroke occurs when blood cannot reach specific areas of the brain often caused by blocked or burst arteries, resulting the death of brain cells. Depending on where the stroke occurs and severity, it can result in different effects.

1.1.1 Key Biomechanical Deficits in hand from strokes

- Reduced grip strength
- Loss of finger individuation
- Flexor synergy patterns
- Reduced finger extension range
- Slower movement speed

In addition to the effects on the patients' hands, other effects that should be considered include:

- Full paralysis on one side of the body
- Vision problems
- Speech/language problems
- Cognitive problems
- Memory problems
- Increased or dulled sensory changes
- Complementary issues like fragile joints, bones and skin

2.1.1 Stages of Stroke Rehabilitation

Table 2.1.2. Timeline with corresponding characteristics of stroke hand recovery.

Stage	Timeline	Biomechanical Characteristics
Acute	0–4 weeks	Flaccid paralysis, minimal voluntary contraction, very low grip force, limited joint control. After a stroke, patient get put in acute care and get assed if suitable for recovery. If they are suitable, they go to rehabilitation at a hospital for 1-3 months.
Subacute	1–6 months	Rapid neural plasticity, increasing voluntary muscle activation, improving grasp and release.
Chronic	>6 months	Recovery slows, compensatory movement strategies appear, persistent weakness often remains.

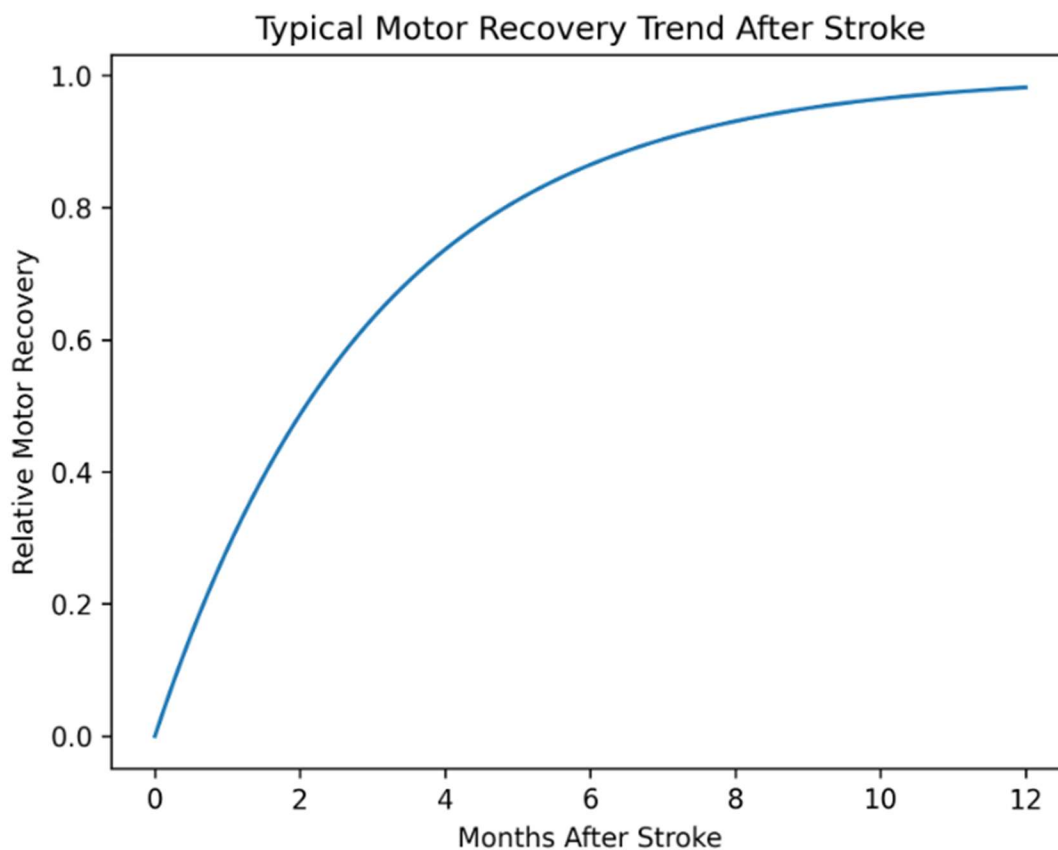


Figure 2.1.2 Motor recovery graph after stroke. Motor recovery occurs most rapidly during the first three to six months following stroke. Neural plasticity during this period allows the brain to reorganize motor pathways and partially restore voluntary movement. After this 6-month period, recovery progress decreases.

2.1.2 Rehabilitation Exercises

Table 2.1.3. Examples of stroke rehabilitation exercises

Exercise	Expected Force	Biomechanics
Grip Training (Therapy Ball/Putty)	50–200 N 300-500 N (healthy)	Finger flexion with wrist stabilization
Finger Extension Training	10–40 N	Opening hand and overcoming flexor spasticity
Pinch Grip Training	10–70 N	Thumb opposition and fine motor control

2.1.3 Struggles with stroke rehabilitation

To understand more about the experiences and struggles of stroke recovery, the team had a meeting with Dr Peter Stubbs, a physiotherapist and researcher at UTS about his experience.

Some key findings from that discussion are:

- Various functional test exists to validate a patient's progress like a pick and place game which patients find more “rewarding” as compared the directly measuring joint angles and spasticity. The measurement of a patient's recovery can only officially be done by their doctor or nurse.
- If patients do not go through their full range of motion, muscles will shorten and patients will lose that range permanently over time. Patients are encouraged to stretch their muscles on their own or with assistance to prevent atrophy.
- The benefit of home rehab is that patients don't have to travel to a physio which can be difficult for the impaired. **Problem is it's difficult to get patients to do rehab at home as they either forget, they're not motivated enough, they can't physically do it or they do their exercises incorrectly.**
- Current issues with virtual gloves are troubleshooting technical issues, input lag, ease of putting on and use, not being able to accurately capture the patient's hand movement.
- Gamification of recovery progress may not directly improve patient's recovery progress but has shown to improve patient's engagement and motivation.

2.1.4 Hand Anatomy

The hand has various joints and muscles with specific names useful when discussing hand movement.

Hand Joints:

Metacarpophalangeal Joint (MCP/MP Joint): The “knuckle” joint at the base of the fingers.

Proximal Interphalangeal Joint (PIP): The first joint in the finger, located below the top two bones.

Distal Interphalangeal Joint (DIP): The joint nearest the fingertip.

Thumb Interphalangeal Joint (IP): The single joint in the thumb.

Carpometacarpal Joint (CMC): Located at the base of the palm, where the metacarpals meet the carpal bones (wrist)

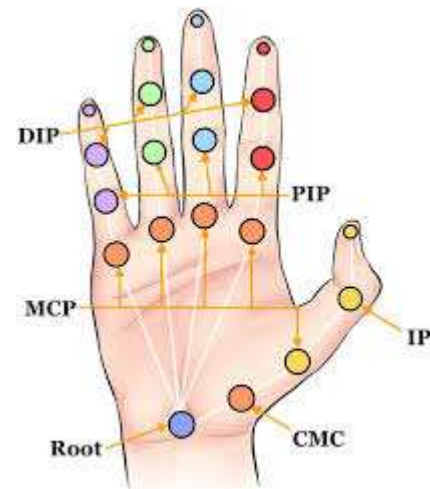


Figure 2.1.5.i. Schematic of individual finger joints

Hand Muscles:

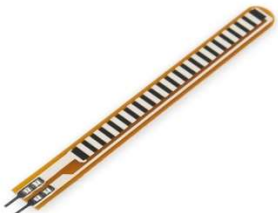
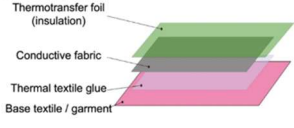
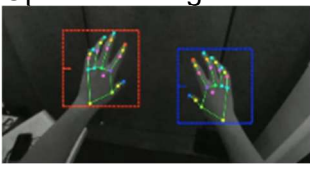
Within the hand itself are the abductor and adduction muscles for the fingers. The flexors and extensor muscles for the hand and fingers are located within the forearm and wrist and connect to the finger joints via tendons. The exception to this is the thumb, which contains a flexor muscle within the hand to flex the MCP joint.



Figure 2.1.5.ii. Schematics of the muscles in the hand (left) and forearm (right).

2.2 Research of Sensors

Table 2.2. Summary of sensor research

Sensors	Sensor function	Pros	Con
Flex sensor 	Measures finger bending through changes in electrical resistance. Used to estimate finger flexion angles.	• Lightweight • Easy to integrate • Simple to use	• Requires calibration • Limited bending range (~90 degree) • Measures flexion only
Inertial Measurement Unit (IMU) 	Inertial measurement unit (IMU is a device that can measure force, angular rate and orientation. Can be used to measure the angle and pose of the hand	• Lightweight • Measures full hand motion • Wide range of motion	• Complex calculations and sensor fusion • Expensive
Capacitive textile patches 	Textile-based capacitive sensors that change capacitance when stretched or deformed. Can be used to identify hand movements and gestures.	• Fully e-textile	• Low angle accuracy • Requires custom fabrication • Requires machine learning processing
Force sensing resistor (FSR) 	Measures applied force through a decrease in resistance as pressure increases. Used to measure fingertip force.	• Small, lightweight and minimal power draw • Flexible • Easy to use	• Requires calibration • Limited range of force sensing (0 – 10N)
Optical sensing 	Uses cameras and computer vision algorithms to estimate hand position and finger movements.	• Can be used with existing camera • Already integrated into other products	• Requires hand visibility • Sensitive to occlusion • Higher processing delay (~200 ms)

2.3 Design selection

After performing research and communicating with stakeholders to understand their needs, the team created a list of potential solutions based and evaluated them using a design matrix as seen below:

Criteria	Weight (%)	Concept 1: Glove with conductive yarn and sensors		Concept 2: Gamification and Community		Concept 3: Virtual hand tracking		Concept 4: Single Finger wearable		Concept 5: Sensor Mat		Concept 6: EMG Tracking	
		Score (1–5)	Weighted Score	Score (1–5)	Weighted Score	Score (1–5)	Weighted Score	Score (1–5)	Weighted Score	Score (1–5)	Weighted Score	Score (1–5)	Weighted Score
E-textile integration (Frederique & Marc)	20	5	1	0	0	0	0	5	1	4	0.8	3	0.6
Gamification/Community (Dr. Stubbs)	10	3	0.3	5	0.5	4	0.4	3	0.3	4	0.4	3	0.3
Comfortability (Frederique & Dr. Stubbs)	15	3.5	0.525	5	0.75	5	0.75	4	0.6	5	0.75	2.5	0.375
Ease of use (Dr. Stubbs)	15	3	0.45	4	0.6	2	0.3	4	0.6	5	0.75	1	0.15
Durability (Frederique)	10	3	0.3	5	0.5	5	0.5	2	0.2	3.5	0.35	2	0.2
Budget (Marc)	5	2	0.1	4	0.2	2	0.1	2.5	0.125	3	0.15	2.5	0.125
Technical Feasibility (TextileTech)	25	2.5	0.625	3	0.75	2	0.5	3.5	0.875	4	1	1	0.25
Weighted Total	100	22	3.3	26	3.3	20	2.55	24	3.7	28.5	4.2	15	2

Figure 2.3. Completed design matrix with our initial concepts.

The design matrix incorporates the needs of the stakeholders as criteria and weighs them based on their overall importance. Each potential solution is then scored on how well it fulfills each. The team initially found a sensor mat idea to be the best solution as we thought it was the most technically feasible solution at the time given the small amount of experience with textiles and the little time we had to produce the solution.

However, after meeting with Frederique to present our solutions, she preferred the wearable solutions more as she also wanted to measure finger joint angles, something which would be difficult to do with a flat mat. In addition, we also met with Shirley, the knitting machine technician who showed us that producing a glove on the knitting machine would be significantly easier than we initially thought.

We concluded on the sensor glove solution for this project, as it was more aligned with what our main client Frederique wanted and now seemed achievable within the time frame with assistance from Shirley.

3. Technical Details

3.1 System Architecture

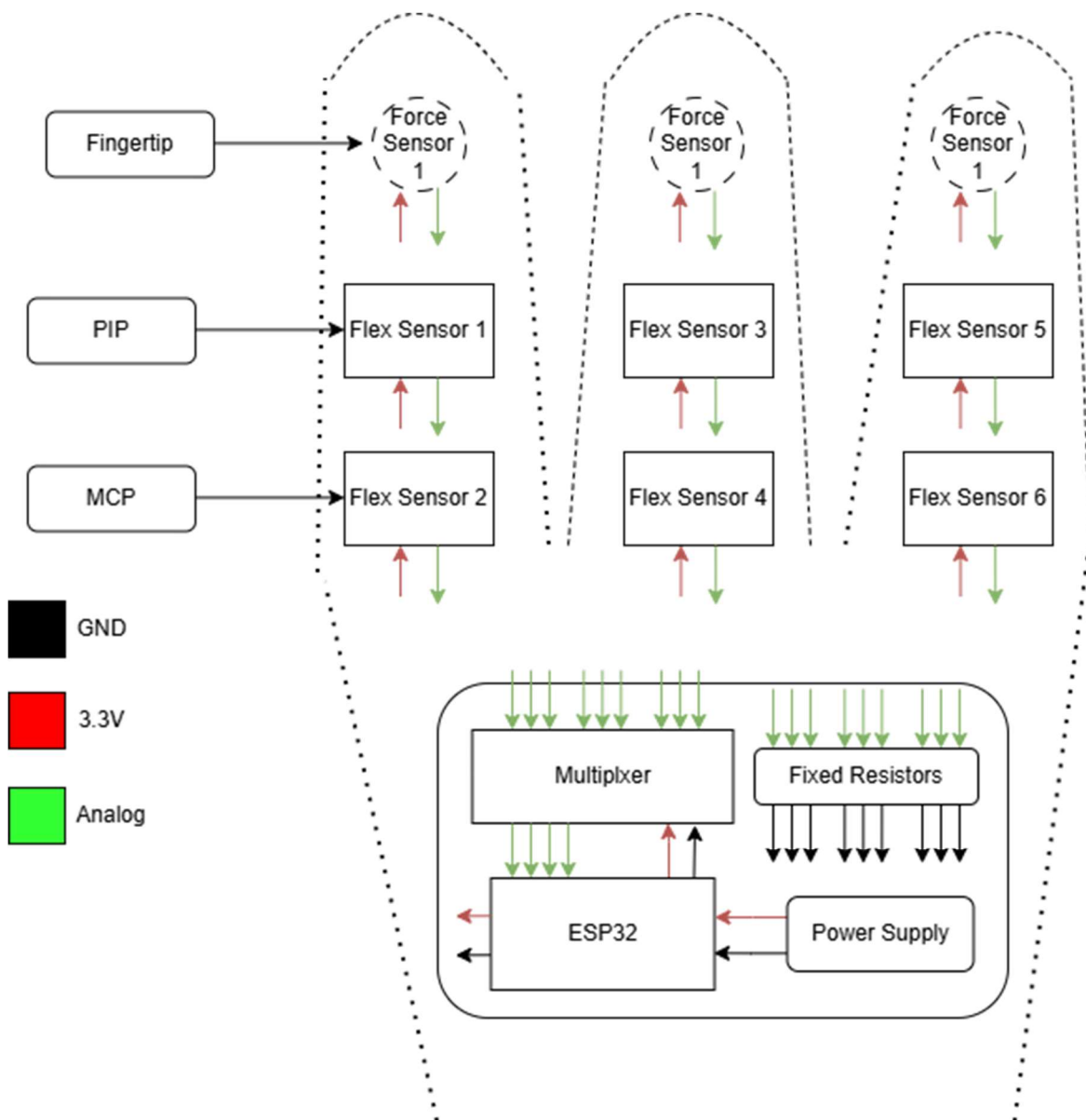


Figure 3.1. Schematic of system architecture

The glove system consists of nine sensors, including six flex sensors for finger joint angle measurement and three force sensing resistors (FSRs) for fingertip force measurement. As shown in the figure, the glove requires a controller shield board where the ESP32 microcontroller is connected to a multiplexer for analog signal selection, as the ESP32 does not have enough ADC pins to directly support all sensors simultaneously. Although the use of a 16-channel analog multiplexer may introduce a small delay, it allows all sensors to be monitored using a single microcontroller system.

The controller board is powered using a 5V power supply. The ESP32 contains an onboard voltage regulator which converts 5V to 3.3V, supplying power to the sensors and multiplexer. Both the flex sensors and FSR sensors require voltage divider circuits for analog signal measurement; therefore,

fixed resistors are integrated into the PCB design (50 k Ω resistors for the flex sensors and 5 k Ω resistors for the FSR sensors).

To improve user comfort and wearability, the glove design minimizes the use of external wiring by integrating conductive yarn traces directly into the textile structure of the glove. Flex sensors are inserted into custom-designed pockets positioned along the PIP and MCP joints to measure finger flexion. Similarly, the FSR sensors are placed into fitted fingertip pockets to measure applied force. However, because the FSR sensors are covered by textile material, the yarn structure may affect sensing performance. Further investigation and optimization will therefore be required in future prototype iterations.

3.2 Requirements

3.2.1 Functional requirements

Requirement No	Requirement
REQ-FUNC_001	The system shall measure finger joint movement during rehabilitation exercises using flex sensors.
REQ-FUNC_002	The system shall measure fingertip force during rehabilitation tasks using force sensing resistors (FSRs).
REQ-FUNC_003	The system shall integrate conductive yarn traces into the glove to reduce the use of external wiring.
REQ-FUNC_004	The system shall provide real-time monitoring and display of rehabilitation sensor data.
REQ-FUNC_005	The system shall support wearable and home-based rehabilitation through a lightweight and comfortable glove design.

3.2.2 Mechanical requirements

Requirement No	Requirement
REQ-MEC_001	During rehabilitation exercises, the system shall detect angular changes in finger position of $\geq 20^\circ$.
REQ-MEC_002	When worn, the system shall maintain sensor alignment within ± 5 mm of intended finger joint positions.
REQ-MEC_003	During rehabilitation tasks, the system shall allow execution of functional hand movements, including grasping, pinching, and finger extension.

3.2.3 Electrical Requirements

Requirement No	Requirement
REQ-ELE-001	The system shall support ≥ 3 analog input channels for sensor data acquisition, including ≥ 1 flex sensor, ≥ 1 pressure sensor, and ≥ 1 spare channel.
REQ-ELE-002	The system shall use an ADC with a resolution of ≥ 12 bits to ensure sufficient sensitivity for detecting small finger movements.

REQ-ELE-003	The system shall measure input voltages within the range of 0–3.3 V with an accuracy of $\pm 2\%$ of full scale.
REQ-ELE-004	The system shall achieve a minimum sampling rate of ≥ 30 Hz per channel for real-time motion tracking.
REQ-ELE-005	The system shall include signal conditioning circuits to convert sensor outputs into a measurable voltage range compatible with the ADC.
REQ-ELE-006	The system shall amplify low-level sensor signals using an operational amplifier to utilize at least 80% of the ADC input range.
REQ-ELE-007	The system shall operate using a portable battery power supply suitable for home use.
REQ-ELE-008	The system shall regulate voltage to provide stable outputs of 3.3 V for MCU and 5 V for sensors (if required).
REQ-ELE-009	The system shall transmit sensor data to an external device via USB, Bluetooth, WiFi, or Serial communication.

3.2.4 Software requirements

Requirement No	Requirement
REQ-SOFT_001	Once the software is launched, the software must automatically search and connect to the sensor hardware.
REQ-SOFT_002	If the software is connected to the sensor hardware, the software must continuously read data from the sensors.
REQ-SOFT_003	If the glove cannot connect to the sensor hardware, it must report a connection error to the user
REQ-SOFT_004	When the software receives raw sensor reading, it must convert the data into meaningful information about the user's hand (e.g forces, angles,scores)
REQ-SOFT_005	When the software receives sensor readings, it must display the readings for the user to help troubleshooting as text
REQ-SOFT_006	When the software converts the sensor readings into meaningful information, it must display the meaningful information as text
REQ-SOFT_007	When the software reads the sensor data, it must also visually display the data as a line plot over time

REQ-SOFT_008	The software must allow user to start/stop recording sensor data
REQ-SOFT_009	The software must allow users to save recorded sensor data in CSV format with time stamps.

3.3 Design Decisions & Rationale

The development process for the rehabilitation glove was divided into multiple prototype stages to reduce technical risk and progressively validate the system design. The first stage focused on independently testing the functionality, durability, and accuracy of the sensors, including flex sensors and force sensing resistors (FSRs). This phase allowed the team to calibrate the sensors, evaluate signal quality, and confirm that the sensors could reliably measure finger movement and fingertip force during rehabilitation exercises.

After validating the sensors individually, the components were integrated into a standard sports glove prototype without conductive yarn. This alpha prototype provided a low-risk platform for evaluating sensor placement, glove mechanics, comfort, and overall electronic integration using conventional wiring. The sports glove was also used to test the ESP32 microcontroller, voltage divider circuits, signal acquisition, and data visualization before progressing to a more advanced textile-integrated design.

Following successful testing of the sports glove prototype, the next phase of development focuses on integrating conductive yarn directly into the glove structure to replace traditional physical wiring. Conductive textile traces were selected because they improve flexibility, comfort, and wearability by reducing bulky external wires that may restrict movement or create discomfort during rehabilitation exercises. Integrating conductive yarn into the textile structure also creates a cleaner and more ergonomic wearable design suitable for long-term use.

The glove was designed to remain lightweight, with a target weight below 150 g, to minimize user fatigue and improve comfort during extended rehabilitation sessions. Since stroke patients may experience muscle weakness and reduced endurance, a lightweight wearable design helps ensure that the glove does not interfere with natural hand movement or discourage repeated use.

The project specifically focuses on home-based rehabilitation because many stroke patients experience reduced rehabilitation progress after leaving supervised clinical environments. Home rehabilitation can be challenging due to limited monitoring, reduced motivation, and incorrect exercise execution. The sensor-integrated glove aims to support rehabilitation in a home setting by providing movement and force tracking, allowing users to monitor their exercises and rehabilitation progress more consistently.

4. Prototype Plan and Testing Plan

The prototype and testing plan is divided into two development stages: Alpha and Beta. The Alpha phase focuses on validating the functionality, accuracy, flexibility, and comfort of the sensors, as well as testing the electrical connections, signal processing, and sensor response time using conventional wiring. Initial testing includes attaching individual flex sensors to the PIP and MCP finger joints using

Velcro straps to evaluate flexibility and bending performance, while individual FSR sensors are attached to the fingertip to test force sensitivity and measurement accuracy. Additional testing is conducted to verify analog signal response, communication stability, and ensure minimal delay during data acquisition.

Following the individual sensor tests, an Alpha sports glove prototype with physical wiring was developed. The sports glove includes sensor pockets on two fingers, consisting of two flex sensors mounted along the PIP joints of the index and middle fingers, and two FSR sensors positioned at the fingertips. This prototype is important because it allows the team to evaluate sensor integration within a wearable glove environment, including sensor placement, glove mechanics, signal quality, and overall system functionality. Sensor data is displayed through a GUI developed using Python, while the ESP32 microcontroller is programmed using PlatformIO. Sensor calibration is also implemented during this stage to improve the accuracy of angle and force measurements.

After completing the Alpha testing phase, the project progresses to the Beta phase, which focuses on integrating conductive yarn into the glove structure to replace conventional wiring. The first Beta prototype consists of a single-finger conductive glove containing two flex sensors and one FSR sensor integrated into textile-based sensor pockets. This simplified prototype is used to evaluate conductive yarn integration, electrical connectivity, durability, and sensor performance within a textile environment before progressing to the final design.

The final prototype stage will be a three-finger conductive textile glove containing six flex sensors and three FSR sensors, as described in the system architecture. For both the Alpha sports glove and Beta conductive glove prototypes, testing is conducted at 0°, 45°, and 90° finger positions to evaluate sensor accuracy, repeatability, and overall rehabilitation tracking performance.

4.1 Individual Sensor Testing

4.1.1 Flex sensor

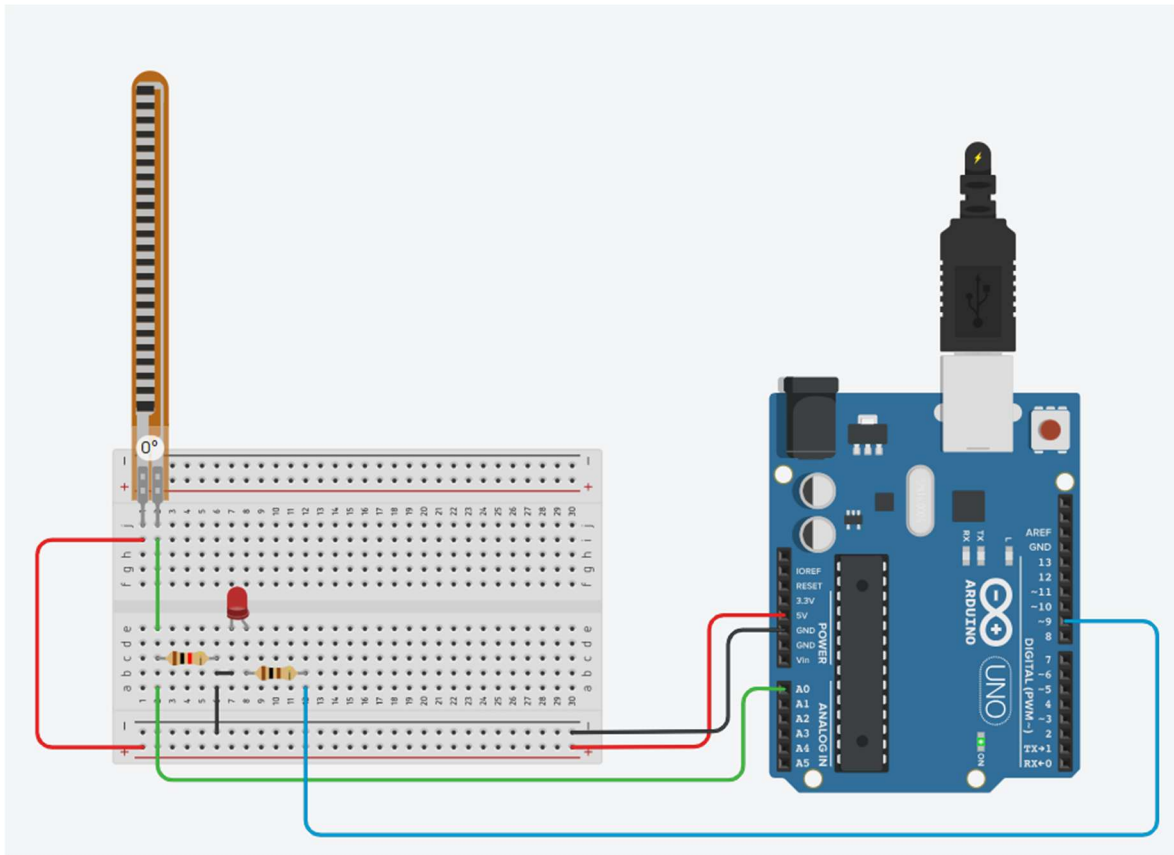


Figure 4.1.1.i. Flex sensor mock circuit with voltage divider

Flexibility test: The client required the glove to measure finger joint angles, which can be achieved using flex sensors. The purpose of this test was to determine whether the flex sensors could bend and follow finger joint movement without causing plastic deformation or damage to the sensors.

The test was conducted by attaching the flex sensors to the MCP and PIP joints using cable ties and moving the finger joints through their full range of motion.

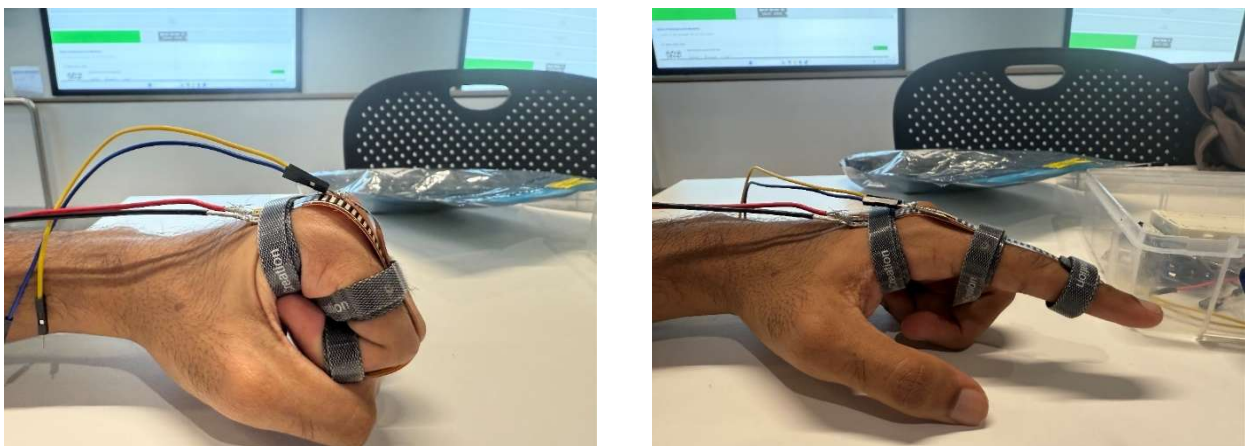


Figure 4.1.1.ii Image of flexibility test with flex sensor on bent finger (left) and straight finger (right)

Results: Flex sensors can match finger joint movements without damage or plastic deformation due to the natural curvature of the joints. However bending flex sensor in sharp, acute angles can cause plastic deformation of sensors.

Accuracy Test: Test how accurately the flex sensor can match finger joint angles by attaching sensors to the index finger and bending fingers at 5 and 10 degrees using 3d printed angled slopes. *Figure 3-vi* shows the experimental circuit which used an Arduino to read and convert the sensor output. The sensor is calibrated by measuring the resistance of the sensor at 0 degrees and 90 degrees and then programming a linear relationship between sensor resistance and angle.



Figure 4.1.1.iii. Image of accuracy test test with flex sensor on a 5 degree fixed angle (left) and 10 degree fixed angle (right)

Results: The flex sensor was able to sense both angles with an average error of 1.5 degrees once calibrated which is acceptable. A problem that was discovered was that the resistance of each flex sensor can vary greatly (~40% difference between the “best” and “worst” sensor we received from our batch) which meant that the sensors had to be calibrated and programmed individually. The flex sensor can also only measure flexion in one direction. Bending the sensor in the other direction will result in unreliable and random readings.

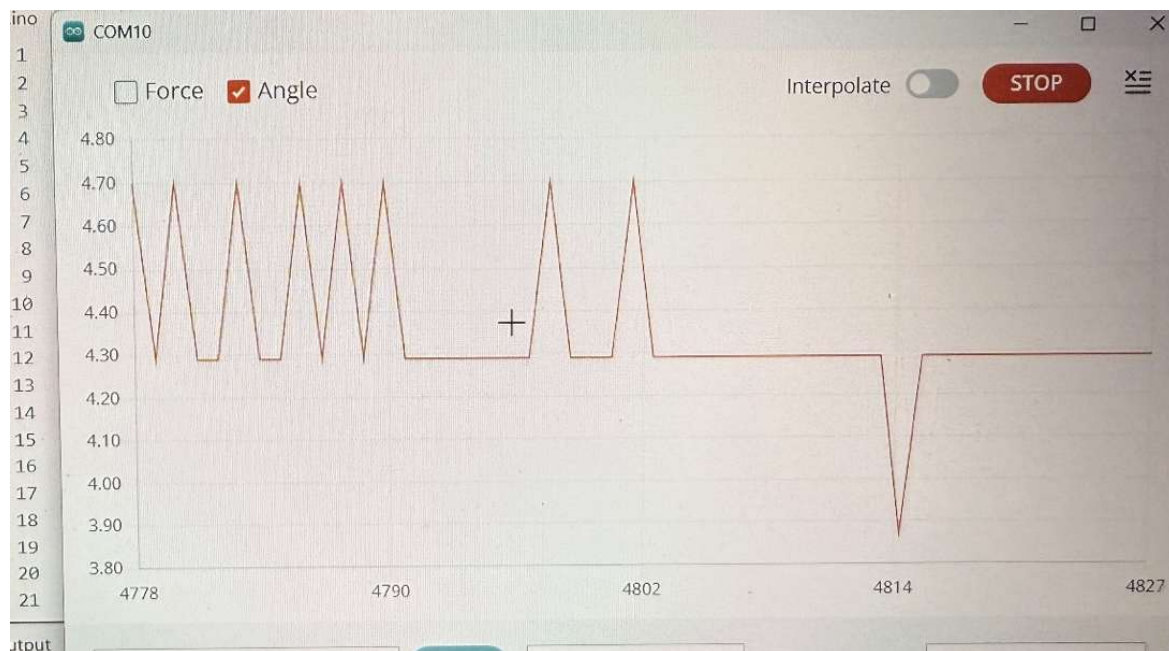


Figure 4.1.1.iv. Screenshoot of v-line graph readings from flex sensor on 5 degree fixed angle

4.1.2 FSR

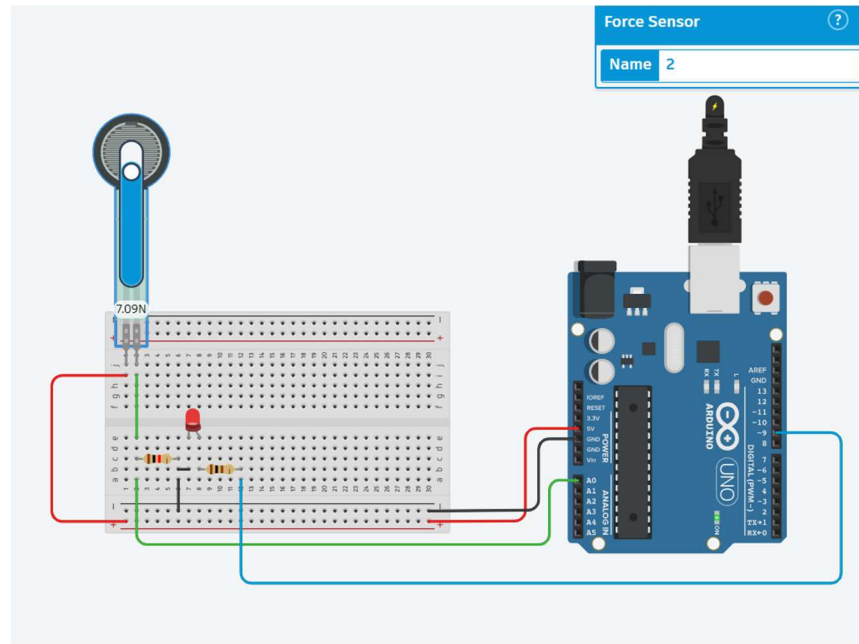


Figure 4.1.2. FSR mock circuit with voltage divider

Accuracy test: Clients wanted gloves to be able to measure the amount of force users can apply at the fingertip, which FSR can measure. Tested how accurately the force sensor we received could measure force inputs. The force sensor was connected to an Arduino to read the sensor output with a circuit seen in *figure 3-vii*.

The sensor was calibrated by first measuring the resistance when no force was applied to the sensor. Afterwards the sensor was placed on an electronic scale, and force was gradually applied by hand until the FSR produced its maximum resistance. This force was recorded and used for the maximum force for the sensor calibration in Arduino programming. To test how accurate the sensor was, the sensor was placed on an electronic scale and force was applied in 200g intervals (200,400,600,800,1000) by pressing down on the sensor by hand and comparing the reading on the electric scale with the sensors reading. This was repeated for all 3 FSR we received.

Result: The average error for all the sensors was 0.08N, which is within acceptable range of errors. Additionally, all the FSR from the batch showed the same amount of accuracy and maximum resistance unlike the flex sensors, meaning each FSR did not need to be individually programme

4.2 Alpha Prototype – Sport Glove with Wire



Figure 4.2.i Image of sports glove. Including top view (left), bottom view (right), and sensors connected with wires (right)

The Alpha prototype consists of a sports glove integrated with sensor pockets and external wired electronics for preliminary testing and validation. The prototype includes three custom-designed sensor pockets, with two pockets positioned on the backside of the glove to house flex sensors for finger joint angle measurement, and one pocket positioned at the fingertip area to house the FSR sensor for force measurement.

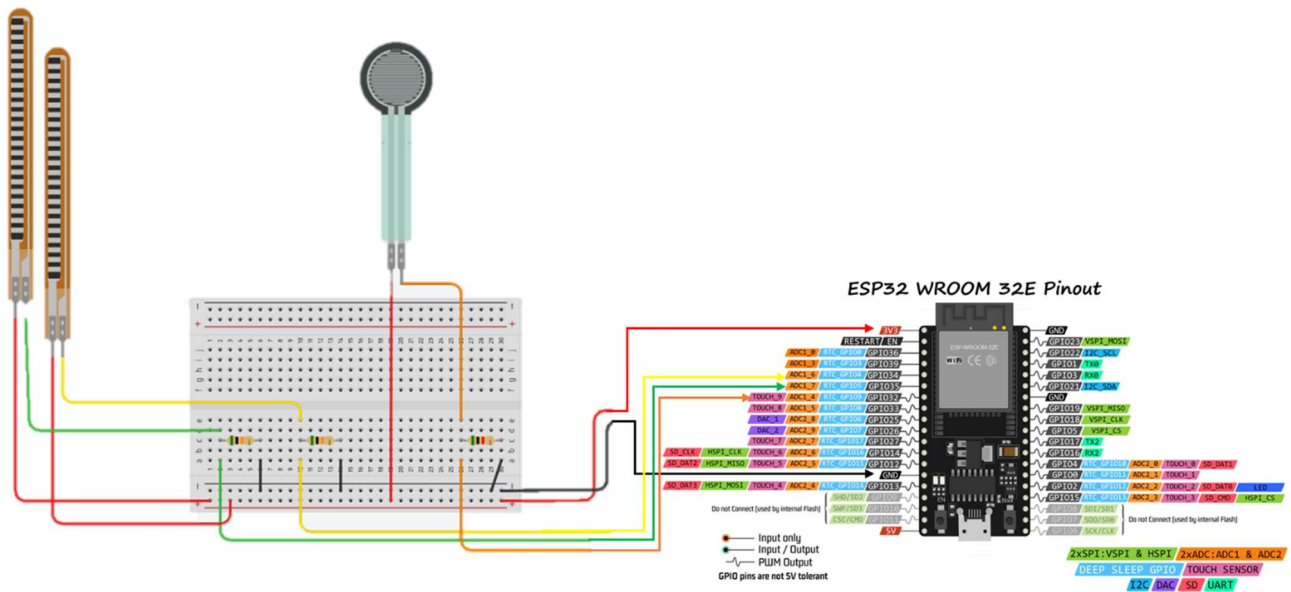


Figure 4.2.ii. Mock circuit of sports glove electronic system

The electronic system consists of an ESP32 microcontroller, a compact breadboard, jumper wire connections, and a mini-USB cable used for powering and programming the ESP32. The sensing system includes two flex sensors for monitoring finger flexion and one force sensing resistor (FSR) for detecting fingertip force during rehabilitation exercises. This prototype was developed to evaluate sensor integration, glove mechanics, signal acquisition, and overall system functionality before progressing to conductive textile integration. Schematic is included in Appendix.

4.2.1 Flex sensor

The first test evaluated the flex sensor response at 0° finger flexion. The results demonstrated that the sensors were able to maintain stable baseline readings with only minor fluctuations caused by electrical noise and the non-ideal testing setup. A goniometer was used as the reference tool for angle measurement during testing. This setup also supported the calibration process of the flex sensors to improve measurement accuracy

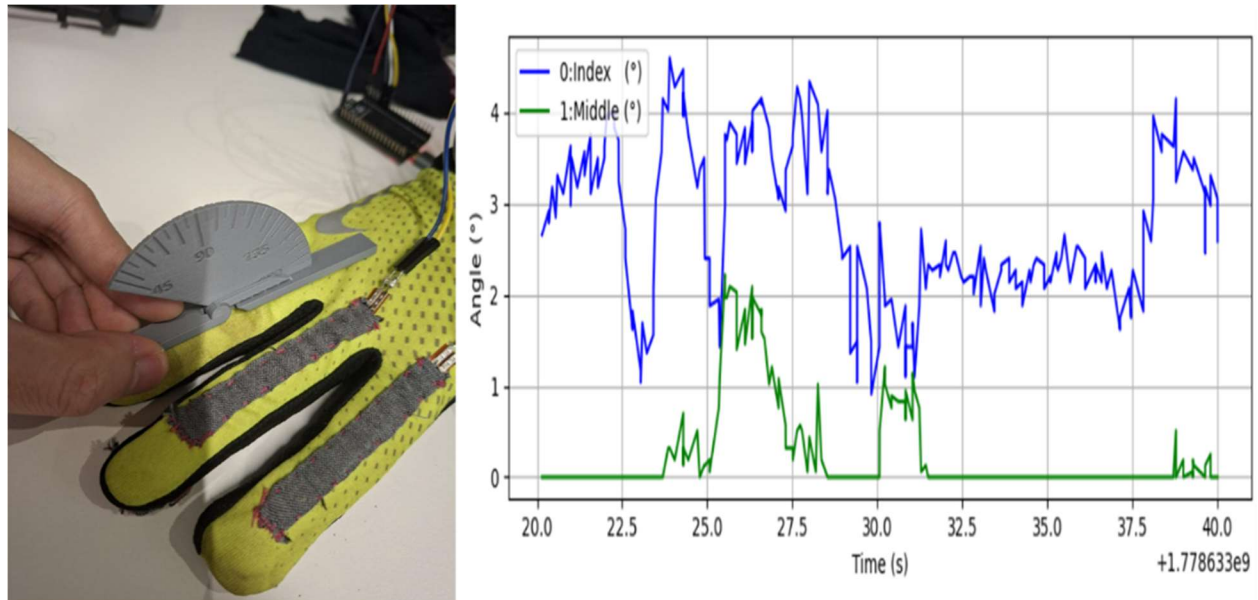
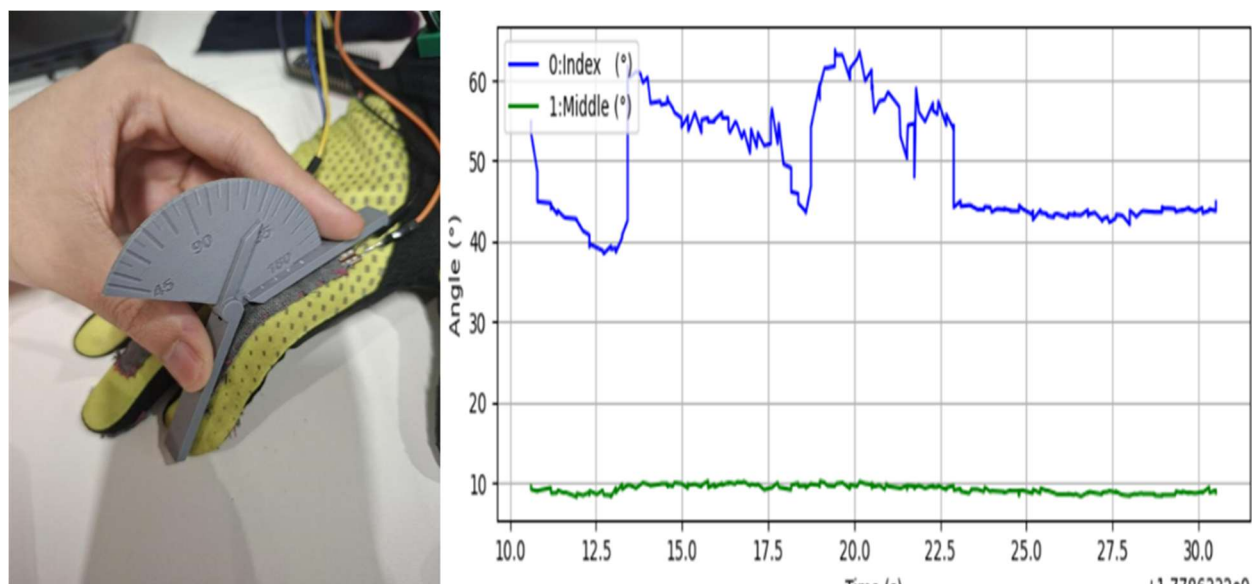


Figure 4.2.1.i Flex sensor results for 0 degrees.

The second test evaluated the sensor response at 45° finger flexion. The results showed high measurement accuracy for both flex sensors positioned on the PIP joints of the index and middle fingers. The sensor outputs clearly reflected the change in finger angle, demonstrating reliable sensor performance during moderate finger bending.



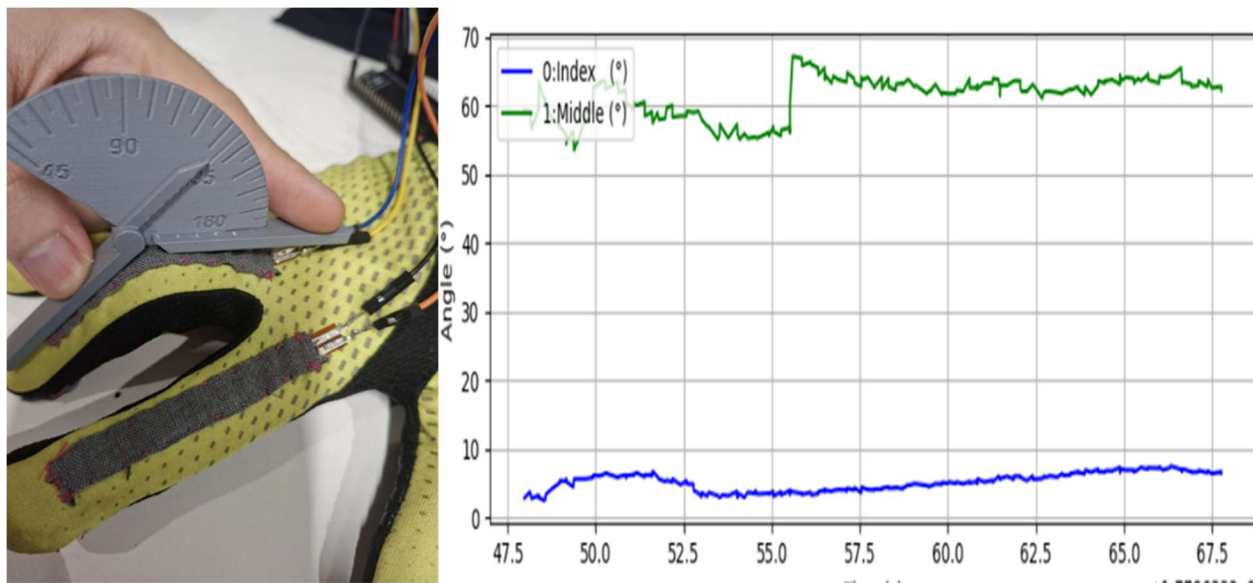
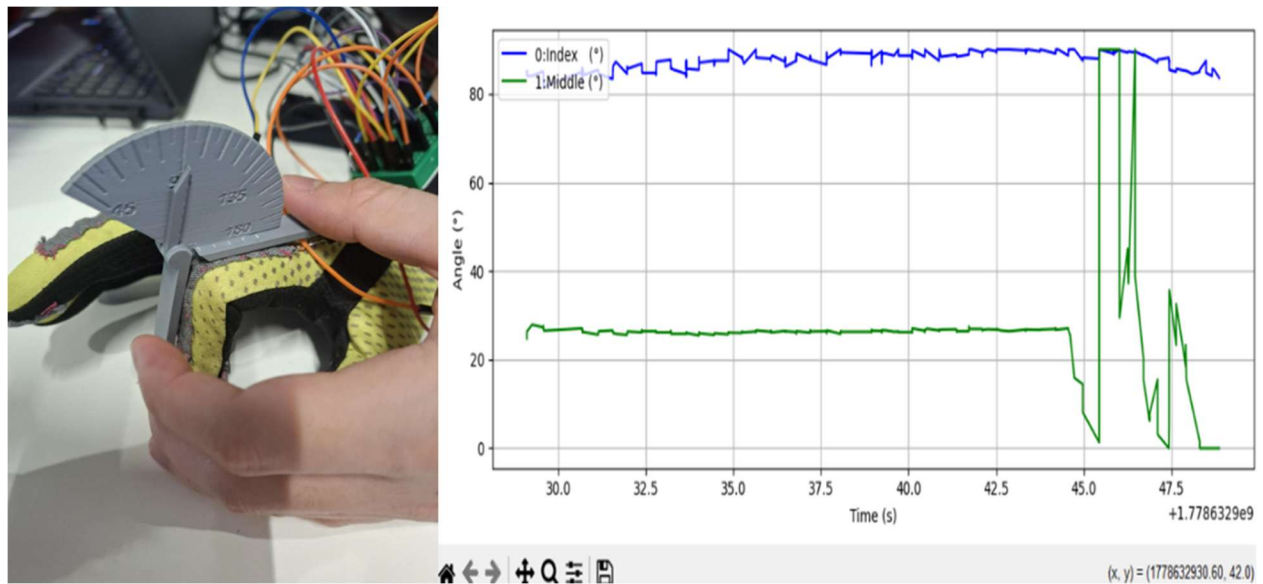


Figure 4.2.1.ii Flex sensor results for 45 degrees on index finger (top) and middle finger (bottom).

The final test evaluated the sensor response at 90° finger flexion. Both flex sensors successfully detected the larger bending angle with clear and distinguishable outputs observed from the plotted graphs. Visual inspection of the data showed consistent sensor behaviour and acceptable performance across the tested angles. These results confirmed that the Alpha prototype achieved the required preliminary functionality and demonstrated that the project was ready to progress into the Beta phase involving conductive textile integration.



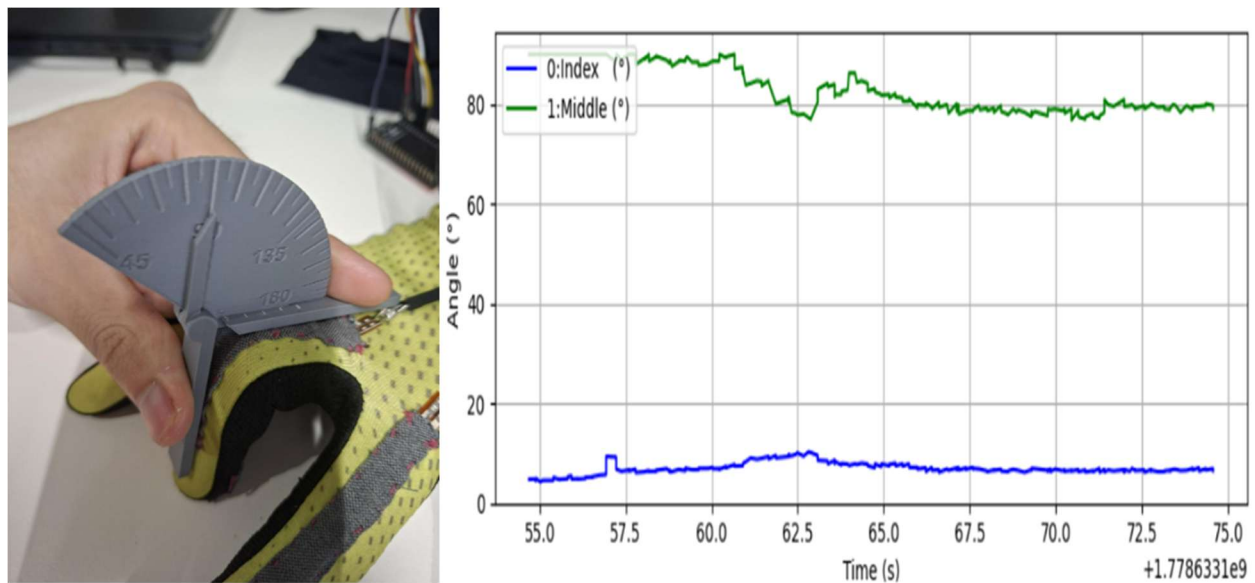


Figure 4.2.1.iii Flex sensor results for 90 degrees on index finger (top) and middle finger (bottom).

4.2.2 FSR sensor

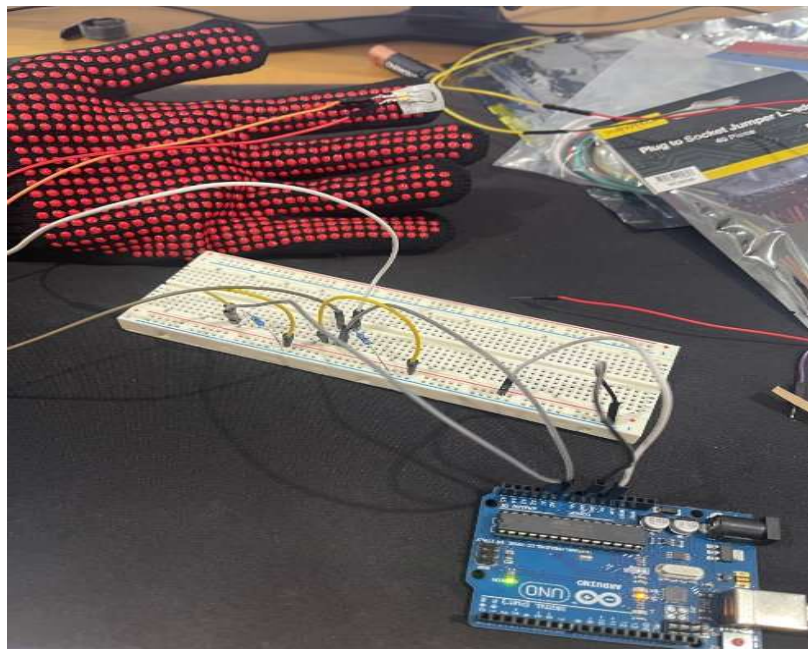


Figure 4.2.2.i Image of FSR Sensor on glove connected to electronics.

During the flex sensor testing phase, one of the FSR sensors was accidentally damaged by the team. However, the electrical setup, circuit design, and sensor integration approach remain unchanged. Further testing and validation of the FSR sensor functionality will therefore be carried out during the Beta phase using the new conductive glove prototype and Arduino-based testing setup.

Preliminary testing of the FSR sensor produced positive results, demonstrating that the sensor was able to respond consistently to increasing applied force. However, due to differences in the current Arduino code implementation, the system was only able to display the general trend of the sensor response rather than calibrated force values. The plotted data shown on the graph represents raw

analog sensor readings transmitted at a baud rate of 9600 using the Serial Plotter tool provided in the Arduino IDE.



Figure 4.2.2.ii Screenshot of Arduino output for FSR sensor.

4.3 Beta Prototype – Single Finger Glove with conductive yarn

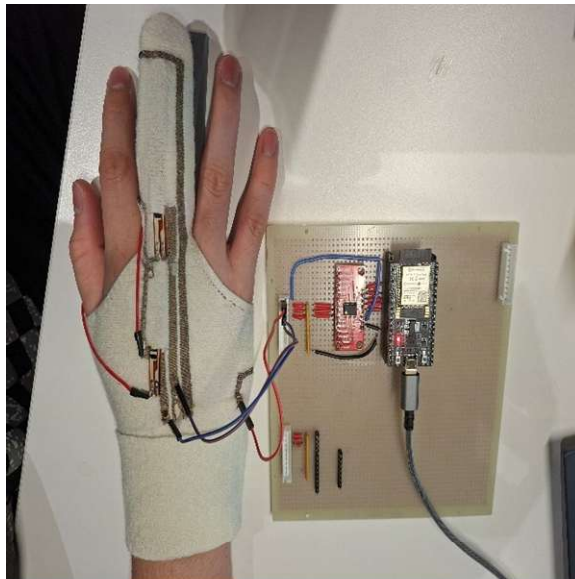


Figure 4.3. Image of single finger glove with conductive yarn beta prototype

This design demonstrates the single-finger conductive glove prototype integrated with conductive yarn and a custom PCB. The main purpose of this phase is to focus on testing the functionality, conductivity, and resistance behaviour of the conductive yarn to ensure that it performs similarly to conventional wiring connections.

Although this setup still requires several jumper wire connections between the conductive yarn traces and the physical electronic components, the flex sensor is directly connected to the conductive yarn using conductive thread integration techniques. Similar to the Alpha sports glove prototype, this testing phase includes electrical continuity and circuit validation to ensure that no short circuits are introduced during the stitching and conductive yarn integration process.

Additional testing was also conducted to evaluate the resistance change of the flex sensor during bending using a multimeter without any powered electrical connection or signal acquisition system. This mechanical-only testing method allowed the team to directly observe the resistance variation of the sensor as the finger bends, helping verify that the conductive yarn integration did not interfere with the natural sensor response.

Following the preliminary electrical and mechanical validation tests, the prototype was evaluated under powered operation by testing sensor accuracy at three different finger angles: 0° , 45° , and 90° , after completing the calibration process. These tests were used to assess signal quality, sensor responsiveness, and the reliability of conductive yarn signal transmission during finger movement. This phase is important for validating conductive textile integration before progressing to the final multi-finger rehabilitation glove prototype.

4.3.1 Conductive Yarn

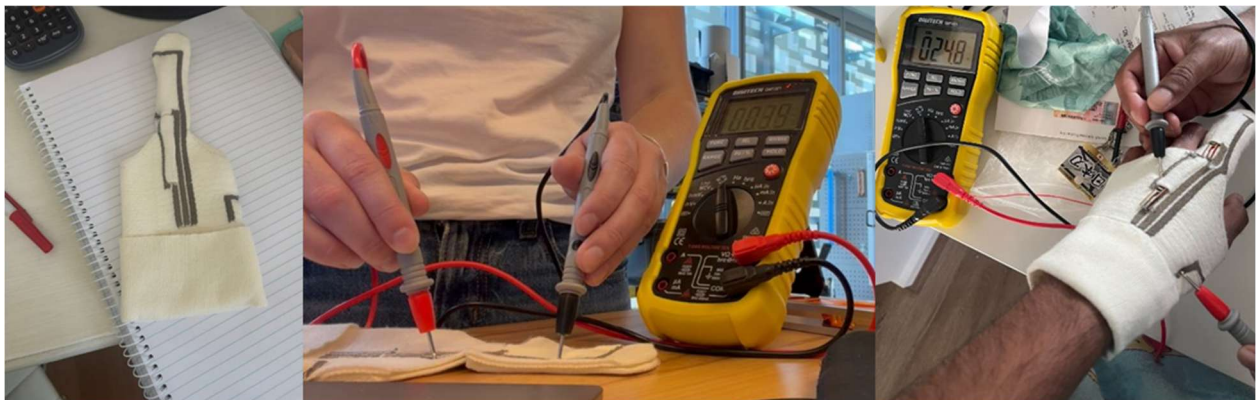


Figure 4.3.1.i Images of testing conductive yarn traces on a single finger glove (left) with a multimeter (middle and right).

The testing of the conductive yarn using a multimeter demonstrated that all stitched conductive pathways were electrically connected correctly and maintained stable continuity throughout the glove structure. Resistance measurements confirmed that the conductive yarn traces were successfully integrated without creating short circuits between neighbouring stitched lines. The results also showed that the conductive yarn maintained low resistance values suitable for analog signal transmission between the sensors and electronic components.

During bending and movement tests, the conductive yarn connections remained stable and did not experience major resistance fluctuations or signal interruption. This demonstrated that the stitching method and conductive yarn integration approach were mechanically reliable for wearable rehabilitation applications. The successful continuity and resistance testing confirmed that the conductive yarn could effectively function as a replacement for conventional physical wiring within the glove system.

A key finding from the bending tests was the relationship between Z-axis trace thickness and signal stability. Increasing the number of yarn layers in the Z-axis (building up the trace vertically) was found

to reduce resistance variation when the trace is bent. This makes thicker Z-axis traces better suited for routing across finger joints, where bending-induced resistance change would otherwise introduce noise. A practical benefit of this is that thicker Z-axis traces can be made narrower in the X-Y plane, allowing more traces to be routed in closer proximity on the glove without interference between channels. This was applied directly in the design of the three-finger glove.

Additionally, the glove material is made from yarn, which improves comfort, flexibility, and wearability for the user. The soft textile structure also supports natural finger movement and allows the flex sensors to bend smoothly along the finger joints during rehabilitation exercises.

4.3.2 Flex sensor

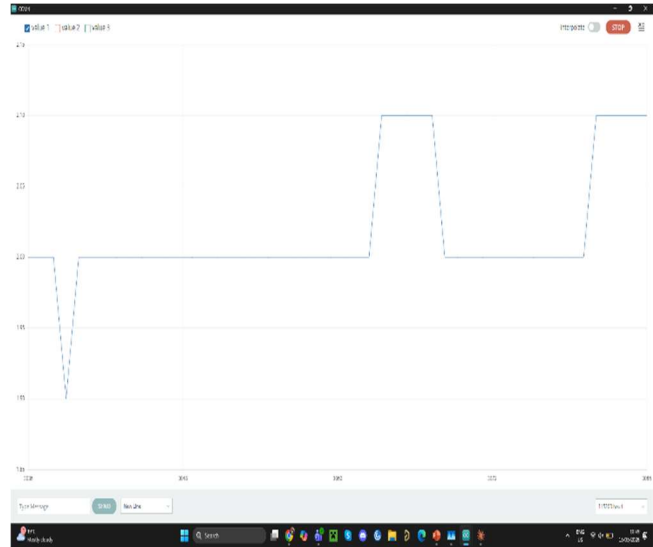
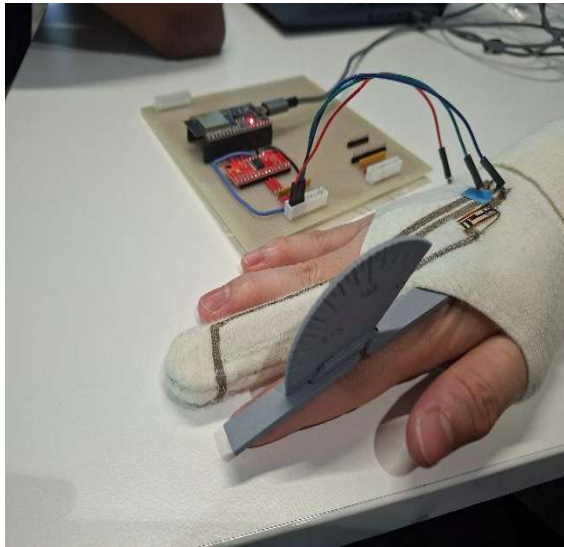


Figure 4.3.2.i. Angle testing 0 degrees on single finger glove with goniometer (left) and sensor output (right).

The testing of the single-finger conductive glove prototype at 0° finger flexion produced stable and reliable results with only minor tolerance observed in the sensor readings. Although small fluctuations were present due to electrical noise and the non-ideal experimental setup, the overall sensor response remained consistent after calibration. Unlike the previous Alpha prototype, the software architecture used in this phase was modified to support the integration of a multiplexer for analog signal selection, as the system now included conductive yarn pathways and additional electrical routing considerations. Due to these hardware and software modifications, the source code for this testing phase was separated and stored in a dedicated GitHub branch named *single_finger_conductive* for version control and future development reference.



Figure 4.3.2.ii. Angle testing 45 degrees on fixed angle support (left) with single finger glove (right).

The second stage of testing evaluated the conductive glove performance at 45° finger flexion. The results demonstrated high measurement accuracy, with an approximate tolerance of $\pm 2^\circ$ after calibration. The plotted sensor outputs clearly reflected the expected change in finger angle, confirming that the conductive yarn integration did not negatively affect sensor performance or electrical connectivity.

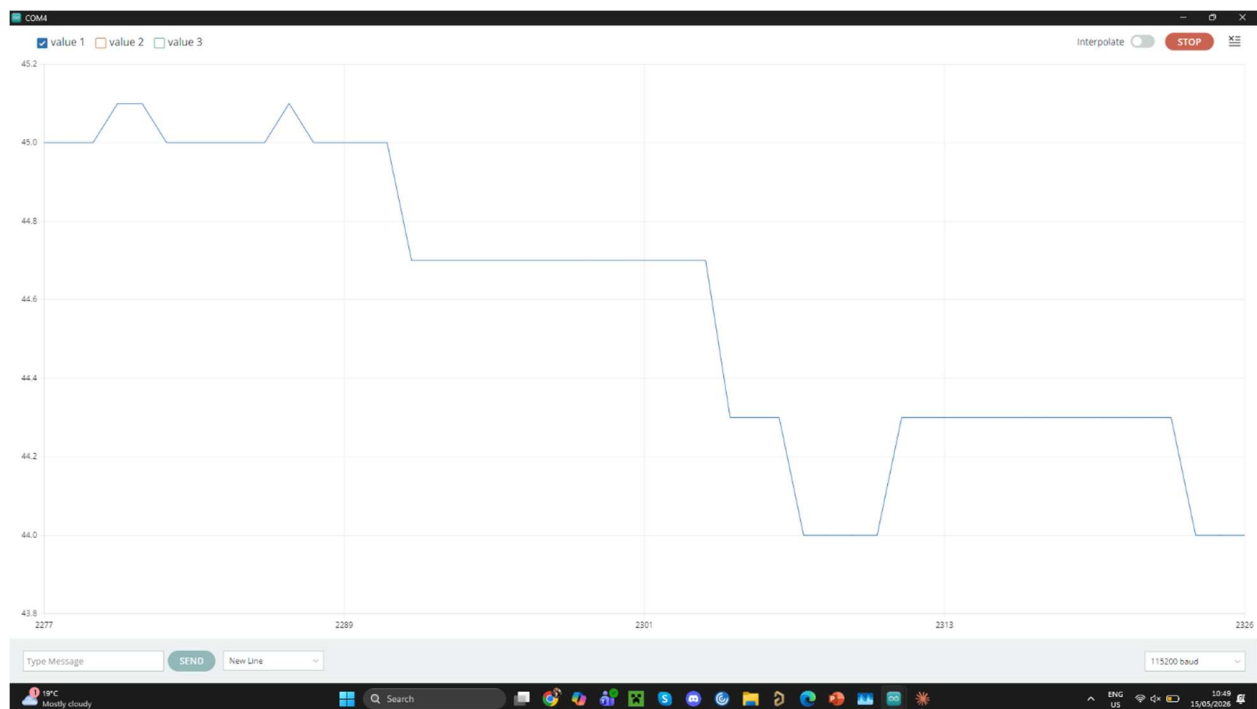


Figure 4.3.2.iii. Sensor outputs of change in finger angle.

This testing phase was particularly important because it validated that conductive textile pathways could successfully transmit analog sensor signals while maintaining reliable communication with the electronic components. The results also demonstrated that the conductive yarn behaved similarly to conventional physical wiring under moderate finger movement conditions.

The final stage of testing evaluated the conductive glove prototype at 90° finger flexion. The sensor outputs closely matched the expected results, with clear and distinguishable graph responses observed during testing. Visual inspection of the plotted data showed stable and repeatable sensor

behaviour across all tested angles. The successful performance of the conductive glove at larger bending angles confirmed that the conductive yarn integration approach was mechanically and electrically suitable for wearable rehabilitation applications. These results demonstrated that the project had successfully validated the conductive textile integration concept and was therefore ready to progress into the final development phase involving the full three-finger conductive rehabilitation glove prototype.

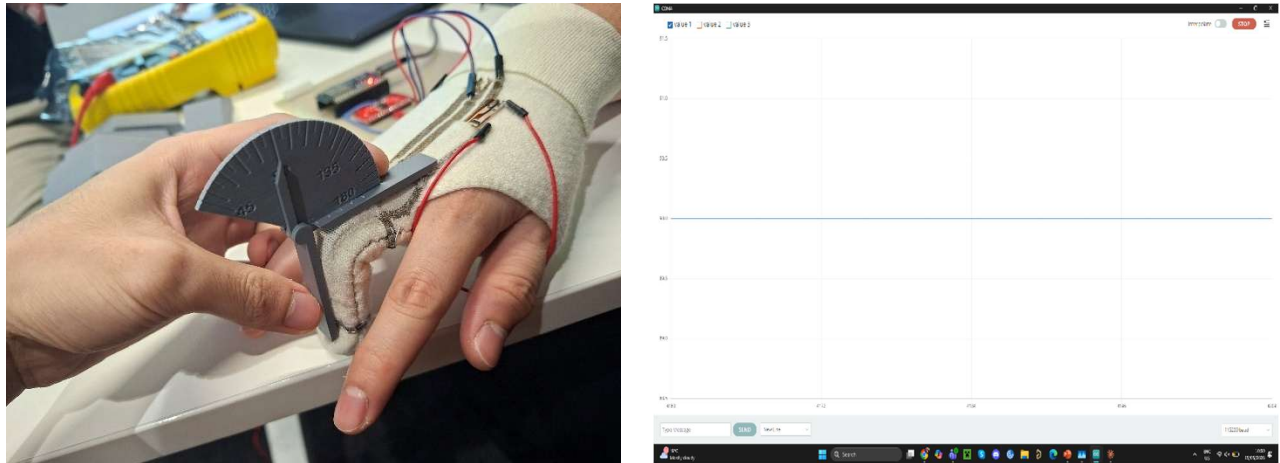


Figure 4.3.2.iv. Angle testing 90 degrees on single finger glove with goniometer (left) and sensor output (right).

4.3.3 FSR sensor

The FSR sensors are positioned at the fingertips of the three-finger glove to detect grip force and contact pressure during rehabilitation exercises. However, full FSR integration testing has not yet been completed, and two issues were identified during initial assessment.

First, the conductive yarn traces on the glove are not currently aligned with the contact pads on the FSR sensors. This means the electrical connections between the traces and the sensor cannot be properly established without adjusting either the trace routing or the sensor placement. This will need to be resolved during the final wiring and assembly stage.

Second, the soft woolly material of the glove pocket housing the FSR appears to disperse applied force across a wider area before it reaches the sensor's active surface. FSR sensors are designed to respond to concentrated point pressure, so the compliant textile layer between the fingertip and the sensor surface may reduce sensitivity and shift the effective measurement range. A firmer or thinner mounting interface may be needed to ensure accurate force readings.

Full FSR integration testing: including threshold calibration and signal validation is pending and will be completed once the sensor mounting and wiring connections on the three-finger glove are finalised.

4.4 Beta Prototype – Final Glove with conductive yarn (3 Fingers)



Figure 4.4. Image of 3 finger glove with conductive yarn

The three-finger conductive yarn glove represents the most advanced prototype developed during this project. The glove integrates conductive yarn traces across three fingers (index, middle, and ring), with each finger designed to accommodate two SpectraSymbol flex sensors at the MCP and PIP joints and one FSR sensor at the fingertip nine sensors in total across the full glove.

The textile structure, knitting, and conductive yarn trace routing are complete. The glove has been physically constructed, and the traces have passed continuity and resistance testing as described in Section 4.3.1. Sensor mounting and wiring connections are not yet finalised the specific challenges encountered during this process are documented in Section 5.

Full-system testing, per-finger calibration, and signal validation are pending and will be the priority for the next team picking up this project. The ESP32 firmware and Python GUI developed during the Alpha and Beta single-finger phases are ready to support this testing with minimal modification.

5. Key Challenges & Limitations

5.1 Sensor Mounting and Textile Integration

One of the most persistent challenges throughout the project was reliably mounting sensors onto the textile glove while maintaining accurate signal output. Multiple approaches were explored, each with its own limitations.

The initial design intended for sensor pockets to hold the sensors snugly in place, with the metal legs of each sensor connecting directly into the conductive yarn traces. In practice this was difficult to execute — the pockets were too loose, allowing sensors to shift laterally and produce inconsistent readings, and the sensor legs were too short and fragile to maintain stable contact with the traces.

Metal snap fasteners were evaluated as an alternative connection method but were ultimately unsuitable. The existing conductive traces on the glove were too narrow to accommodate them without risk of the fasteners overlapping onto adjacent traces and causing short circuits.

Soldering the sensor legs directly onto the conductive yarn was also attempted. While functional connections were achieved, the process was extremely difficult — the heat required for soldering risks melting the yarn, and the yarn lacks the flux properties needed to form a reliable joint, meaning connections could easily be pulled apart. A further issue discovered during this process was that wearing the glove causes the textile to stretch, which shifts the sensor's position relative to the underlying traces. Any rigid connection method therefore risks disconnecting simply from the act of putting the glove on or taking it off, meaning the connection approach needs to accommodate this movement.

Wire glue was trialled as a no-heat alternative, but similarly failed to hold reliably over time. Glove stretching during wearing caused the glue to separate from the trace as the sensor's relative position changed.

The most effective approach was knitting the sensors directly into the glove structure during fabrication. While this worked better than all other methods, the pockets remained somewhat loose in the final design, still allowing occasional lateral sensor movement and contributing to inconsistent readings in some test cases.

A structural consideration for anyone continuing this work is that the current glove requires three layers of textile — a base layer, a conductive yarn layer, and an outer insulating layer — to prevent adjacent conductive traces from short-circuiting when the glove folds or is compressed. This adds thickness and manufacturing complexity that should be considered in future design iterations.

Additionally, the FSR pressure sensors presented their own integration challenges within the textile environment. The conductive traces on the current glove are not fully aligned with the FSR sensor pads, and the soft, flexible material of the knitted glove pocket appears to partially reduce sensor sensitivity by dispersing force across the pocket surface rather than concentrating it onto the sensing area. Resolving the trace alignment and FSR pocket design remains an outstanding task for the final prototype.

5.2 Sensor Calibration

Flex sensor calibration is non-trivial due to the non-linear relationship between sensor resistance and finger bend angle, which is further complicated by electrical noise. Accurate calibration requires measurements at multiple reference angles rather than a simple two-point approach. A further complication is significant variation in resistance values between individual sensors — while the datasheet may specify a typical base resistance, sensors from the same batch were found to have base values ranging approximately 20–30 k Ω and similarly variable maximum values. This means every flex sensor must be calibrated individually. Additionally, prolonged heat exposure can permanently alter these values by deforming the carbon ink layer inside the sensor, which should be kept in mind during testing and storage.

A potential calibration approach identified partway through the project is to record each sensor's resistance values at a set of fixed reference angles once at the beginning of the glove's life. Since these values appear to remain relatively stable over time under normal use conditions, this may reduce the need for ongoing recalibration and could form the basis of a more practical calibration process for a finished product.

FSR pressure sensor calibration behaves differently. The response curve is highly non-linear, with roughly 80% of the sensor's measurable force range covered by what most users would consider a light touch, and the upper 20% requiring considerably more force. This characteristic is actually suitable for the stroke rehabilitation context, where patients typically apply lighter forces, and can be further adjusted by changing the resistance value in the voltage divider circuit if needed.

Calibrating the FSR to specific Newton values requires applying known forces at a concentrated point on the sensor. The team found that placing flat weights such as coins directly onto the sensor underestimated the applied force because the load was dispersed across the sensing surface rather than concentrated. A small rigid spacer between the weight and the sensor was used to concentrate the load, though the sensor occasionally failed to register constant unchanging pressure reliably over time. Despite this, the sensor performed adequately for detecting relative changes in applied force during rehabilitation exercises, which remains the primary use case.

5.3 Conductive Yarn Capabilities and Limitations

The conductive yarn performed well as an electrical signal transmission medium. It successfully carried analog signals from the finger-mounted sensors through to the controller board without significant resistance or signal loss, validating its use as a replacement for external physical wiring and improving the overall comfort and wearability of the glove.

However, the conductive yarn is not suitable as a sensing material. When bent, the yarn produces resistance changes on the order of tens to low hundreds of ohms, which is far below the kiloohm-range changes produced by dedicated flex sensors. This makes the yarn incapable of accurately measuring finger joint angles or applied forces through resistance change alone, and dedicated sensors remain necessary for this purpose.

An important finding from the yarn integration phase relates to the relationship between trace thickness and resistance under bending. Traces that are thicker in the Z-axis — meaning more layers of stitching stacked on top of one another — maintain lower resistance when flexed around finger joints and glove edges, which improves signal quality. This understanding allowed the team to keep traces thinner in their surface footprint (X-Y dimensions), enabling multiple traces to be placed closer

together on the glove without interference. This was a critical space constraint given the number of signal lines the glove requires across three fingers.

5.4 User Variability and Fit

The current glove design is not easily adaptable to different users. Hand size and finger length vary significantly between individuals, meaning the sensor positions — and potentially the glove construction itself — would need to be adjusted or re-knitted for each patient. This presents a practical barrier to scaling the device beyond a single-user research prototype without developing a more flexible sizing system.

In the current prototype, the PCB controller board is also not yet mounted to the user's arm, meaning a short run of external wiring still connects the glove to the electronics. Developing a compact, arm-mounted enclosure for the PCB and power supply would significantly improve the portability and wearability of the system for real rehabilitation use.

5.5 Time and Resource Constraints

As a university semester project, the available time for hardware development and testing was limited. The first six weeks of the project were necessarily spent on research, client consultations, stakeholder meetings, and evaluating the most suitable prototype direction before any physical build work could begin. The remaining time was then shared across ideation, prototyping, testing, and iteration — all while managing concurrent subject workloads. As a result, certain planned testing phases, particularly full three-finger system testing and complete FSR integration within the textile glove, could not be completed within the semester. Future teams or funding bodies continuing this work should factor this context into their understanding of what remains to be validated.

6. Data, Files & Resources

6.1 GitHub Repository

All project code, CAD files, presentation slides, and media are stored in the following GitHub repository:

<https://github.com/Nam2862003/DMMS>

The repository contains the ESP32 firmware (developed using PlatformIO) and the Python-based GUI for real-time sensor data visualisation and recording. The software supports all prototype stages, with glove configuration selected from within the UI. Presentation slides from tutor reviews and progress updates are also stored in the repository, along with CAD files for the 3D printed testing tools (finger angle slopes and goniometer) and a media folder containing screenshots and videos of sensor test results.

6.2 Physical Items Being Handed Over

The following physical items are being handed over alongside this document:

- Alpha prototype — sports glove with conventional wiring, flex sensors, FSR, ESP32, and breadboard assembly
- Beta prototype (single finger) — single-finger conductive yarn glove with custom PCB

- Final prototype — three-finger conductive yarn glove (textile structure and traces complete; sensor and wiring connections to be finalised)
- 3D printed angle testing slopes (10°, 20°, 35°, 40°, 45°, 55°, 80°, 90°)
- 3D printed goniometer
- Remaining components (flex sensors, FSR sensors, conductive yarn, spare PCB components)



Figure 6.2. Image of all items being physically handed over

6.3 Component Datasheets & Technical References

Component	Reference
Flex Sensor	SpectraSymbol Flex Sensor Datasheet
Force Sensing Resistor (FSR)	Interlink Electronics FSR Datasheet
Flex Sensor Breakout Board (BOB-09056)	SparkFun Breakout Board Datasheet
Resistor Network — Voltage Divider (Bourns 4600x)	Bourns 4600x Datasheet
ESP32 WROOM-32E	Espressif ESP32 Datasheet
ESP32 DevKitC V4 Pinout	ESP32 DevKitC V4 Pinout & Specs
Flex Sensor – ESP32 Interfacing Guide	ElectronicWings Guide

Component	Reference
Flex Sensor Hookup Guide	SparkFun Hookup Guide

6.4 Recommended Reading

The following papers are recommended as background reading for anyone continuing this project:

Wearable Rehabilitation Gloves:

- Rehabilitation For Stroke Survivors: The Development of a Smart Glove
- Design and Development of a Stroke Rehabilitation Glove for Measuring and Monitoring Hand

Motions

- Wearable Technologies for Hand Joints Monitoring for Rehabilitation: A Survey
- Wearable Multimodal Sensing-Based Serious Games for Hand Movement Training After Stroke

E-Textiles & Conductive Yarn:

- Analysis of Sensitivity, Linearity, Hysteresis and Fatigue of Textile Knit Stretch Sensors
- Electronic Textiles and Wearable Technologies

7. Recommendations for Future Work:

7.1 Complete the Three-Finger Glove Prototype

The most immediate priority is completing the final three-finger conductive yarn glove prototype. The textile structure and conductive yarn traces are complete; the remaining task is finalising the sensor and wiring connections and then running full system calibration and testing across all three fingers. This should be the first task any continuation team undertakes before progressing to any of the improvements below.

7.2 Improve Sensor Mounting and Connection Reliability

Sensor mounting remains one of the most unresolved challenges in the current design. The knitted pocket approach worked best out of all methods trialled but still leaves sensors with too much lateral movement, affecting reading consistency. A more reliable and repeatable mounting solution needs to be developed. Additionally, the Arduino jumper wires currently used as connectors between the glove and the PCB board can come loose easily during use and should be replaced with more secure, purpose-made connectors that can withstand the mechanical stress of repeated wearing and removing.

The FSR sensor integration also requires specific attention — the conductive traces on the current glove are not yet aligned with the FSR sensor pads, and the soft woolly pocket material appears to reduce sensitivity by dispersing force before it reaches the sensing area. A harder pocket insert or a revised placement approach should be explored to ensure consistent force readings.

7.3 Design a Proper PCB with Wireless Communication

The current electronics setup uses a breadboard and jumper wires, which is appropriate for prototyping but not suitable for a finished rehabilitation device. A properly designed PCB should be developed that integrates the ESP32, multiplexer, voltage divider resistors, and power supply circuitry into a single compact board. The ESP32 supports both Bluetooth and WiFi natively, so transitioning to wireless data transmission would eliminate the USB cable currently required for operation and make the device genuinely portable for home rehabilitation use.

7.4 Develop an Arm-Mounted Enclosure

Alongside the PCB redesign, a compact arm-mounted enclosure should be designed to house the electronics on the user's forearm. This would remove the last remaining external wiring between the glove and the controller board, completing the transition to a fully wearable, wireless rehabilitation device.

7.5 Improve Signal Accuracy with Electrical Filtering

The current sensor readings exhibit noise and minor fluctuations, particularly in the flex sensor outputs. Adding capacitors and low-pass filter circuits to the PCB design would reduce high-frequency electrical noise and improve the stability and accuracy of both flex sensor and FSR readings, making calibration easier and measurement results more reliable.

7.6 Improve the Calibration Process

The current calibration process requires individual calibration of each flex sensor due to significant batch-to-batch resistance variance. A more systematic approach should be developed — specifically, recording each sensor's resistance at a set of fixed reference angles once during initial glove setup, then using those values as a stable calibration baseline. This would reduce the time required for recalibration and make the process more practical for clinical or home use. For FSR calibration, a standardised weight-based approach using a concentrated load point should be formalised and documented so that future teams and end users can repeat it consistently.

7.7 Refine Glove Sizing

The current glove is designed for a single hand size. A system for adapting the glove to different hand sizes and finger lengths should be developed, as stroke patients vary significantly in hand geometry. This could involve designing a range of standard sizes or developing a more adjustable construction approach that allows sensor positions to be shifted without rebuilding the entire glove.

7.8 Develop a Mobile App or Improved Software Interface

The current Python GUI provides real-time data visualisation and CSV export but requires a laptop connection. Developing a mobile application that communicates with the glove over Bluetooth would significantly improve usability in a home rehabilitation setting. The app could display real-time finger movement and force data, track rehabilitation progress over sessions, and potentially provide exercise guidance or feedback to help patients maintain correct form.

7.9 Conduct User Testing with Stroke Patients

No user testing with actual stroke patients has been conducted to date. This is a critical step before the device can be considered for clinical or home use. Testing should evaluate comfort and wearability, ease of putting on and removing the glove independently (particularly important given the motor impairments of the target users), accuracy of measurements during real rehabilitation exercises, and patient engagement with the data feedback.

8. Risks & Considerations

8.1 Safety

Stroke patients frequently experience reduced sensation in the affected hand, meaning they may not notice if the glove is applying uncomfortable pressure or restricting circulation. The glove fit must be monitored carefully, particularly around the fingertips and knuckles where sensor pockets add localised thickness. Additionally, as noted in the stroke rehabilitation research, patients often present with fragile joints, bones, and skin — meaning any rigid or overly constrictive components in the glove design carry a risk of causing discomfort or injury during use. Future design iterations should include fit assessment as part of any user testing protocol.

The electrical system operates at 3.3V and 5V, which poses no significant electrical safety risk to users under normal operating conditions. However, any transition to a battery-powered wireless design should include appropriate protection circuitry to prevent overcharge, short circuit, or thermal issues.

8.2 Durability and Maintenance

The current glove prototype is not washable. Conductive yarn traces and integrated electronic components cannot be exposed to water without risk of damage or permanent resistance change. For a device intended for daily home rehabilitation use, this is a significant practical limitation. Future designs should consider either a removable electronics module that separates from the washable textile component, or the use of waterproof coatings on the conductive traces.

Conductive yarn traces may also degrade over repeated bending and stretching cycles, potentially increasing resistance over time and introducing calibration drift. The durability of the stitched traces under long-term rehabilitation use has not yet been tested and should be evaluated in future work. Flex sensor longevity is also a consideration — the carbon ink layer inside the sensor can degrade under repeated sharp bending or prolonged heat exposure, affecting accuracy over time.

8.3 User Compliance and Accessibility

Research and consultation with Dr. Peter Stubbs highlighted that patient compliance is one of the most significant challenges in home rehabilitation. Patients frequently forget to exercise, lack motivation, or perform exercises incorrectly without supervision. For the glove to be effective in a home setting, it must be simple and quick to put on and take off independently — which is particularly challenging given that the target users have reduced motor function in the affected hand. Any design that requires two functioning hands to set up will significantly limit usability.

Technical issues such as calibration drift, connectivity failures, or software errors could also discourage patients from using the device consistently. The software interface should be designed to minimise troubleshooting requirements and provide clear, simple feedback to users with varying levels of technical literacy.

8.4 Clinical and Regulatory Considerations

The current prototype has not undergone clinical validation and should not be used as a medical diagnostic tool. As noted during consultation with Dr. Stubbs, official measurement of a patient's rehabilitation progress can only be conducted by a qualified clinician. The glove is intended to support and supplement rehabilitation exercises, not to replace clinical assessment.

Any formal user testing with stroke patients will require ethics approval through the relevant institutional review board prior to commencement. If the project progresses toward a commercially available medical device, regulatory approval through the Therapeutic Goods Administration (TGA) in Australia would also be required, which involves a significantly more rigorous validation and documentation process than academic prototype testing.

9. Timeline / Roadmap Forward

9.1 Short-Term — Weeks 1–4 (Complete the Current Prototype)

The priority for any continuation team is completing the three-finger conductive yarn glove prototype to a fully testable state. This includes installing the remaining three FSR sensors at the fingertips of the index, middle, and ring fingers, and integrating the remaining two MCP flex sensors for the middle and ring fingers. This involves finalising the sensor and wiring connections on the glove, replacing the existing Arduino jumper wire connectors with more secure alternatives that can withstand repeated wearing and removal, and aligning the FSR conductive traces with the sensor pads. Once the connections are stable, full system calibration should be performed across all three fingers — covering both flex sensors at the MCP and PIP joints and the FSR sensors at each fingertip — and testing conducted at 0°, 45°, and 90° finger positions, consistent with the methodology used in the Alpha and Beta single-finger prototype testing. The FSR integration within the textile pocket should also be evaluated and documented, completing the outstanding section 4.3.3 of this handover document.

9.2 Medium-Term — Weeks 5–12 (Hardware and Software Improvements)

With a fully tested prototype in hand, the focus should shift to addressing the key hardware limitations identified during this semester. The most impactful improvements to pursue within this phase are designing a proper PCB that consolidates the ESP32, multiplexer, voltage divider resistors, and power supply into a single compact board with wireless communication capability and refining the existing mechanical enclosure into a compact forearm-mounted housing capable of securely containing the PCB, battery, and communication hardware. Together these changes would eliminate the remaining external wiring and produce a genuinely portable, wireless rehabilitation glove. Adding capacitors and low-pass filter circuits to the PCB design during this phase would also meaningfully improve signal accuracy by reducing the electrical noise observed in current sensor readings.

At present, the ESP32 requires a micro-USB connection for both power delivery and communication with the PC GUI. Implementing Wi-Fi or Bluetooth communication should therefore be prioritised to eliminate the tethered connection and improve portability for home rehabilitation use.

Alongside the hardware work, the sensor calibration process should be formalised into a systematic procedure using preset reference angles for each flex sensor, and glove sizing should be revisited to explore whether a range of standard sizes or an adjustable construction approach can be developed to accommodate different hand geometries.

9.3 Long-Term — Beyond the Semester (Clinical Development and Expansion)

Once the hardware is stable and the glove performs reliably across different users, the project is ready to move toward clinical development. This phase begins with ethics approval and formal user testing with stroke patients, evaluating comfort, ease of use, measurement accuracy during real rehabilitation exercises, and patient engagement with the data feedback. Findings from this testing should directly inform further design refinements.

In parallel, a mobile application should be developed that communicates with the glove over Bluetooth, displaying real-time movement and force data, tracking rehabilitation progress across sessions, and providing exercise guidance to support patients exercising independently at home. Incorporating gamification elements — such as progress tracking, goal-setting, and interactive exercises — has been shown to improve patient motivation and engagement in rehabilitation and should be considered as part of the software development roadmap.

If the project progresses toward a deployable medical device, regulatory approval through the Therapeutic Goods Administration (TGA) in Australia will be required, alongside formal clinical validation studies to demonstrate the device's safety and effectiveness in a rehabilitation setting.